

ELEVATORS PROJECT MONITORING DASHBOARD

Excel Data Analytics & Visualization Project

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Tools:

Microsoft Excel | Pivot Tables | Pivot Charts | Slicers | Data Analysis

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1. PROJECT INTRODUCTION

This project presents an interactive Excel-based dashboard developed for monitoring elevator installation projects. The dashboard brings together key project information in a centralized and easy-to-understand format.

It provides an overview of project status, installation stages, handovers, delays, customer types, regions, cities, and project value. This helps users monitor multiple projects efficiently and understand the current performance of ongoing and completed projects.

The dashboard uses Pivot Tables, Pivot Charts, KPI cards, and interactive slicers to transform project data into meaningful visual insights. Users can filter the dashboard based on region, customer type, and project status to analyse the information from different perspectives.

The overall purpose of the dashboard is to support efficient project monitoring, identify areas requiring attention, and provide management with a clear view of project performance for better decision-making.

2. BUSINESS PROBLEM

Managing multiple elevator installation projects requires continuous monitoring of project status, installation stages, delays, handovers, customer types, regions, and project value. When project information is maintained across different records or reviewed manually, it can become difficult to obtain a clear and timely view of overall project performance.

The absence of a centralized monitoring system may make it challenging to identify delayed projects, track pending handovers, compare project performance across regions, and understand the distribution of project value.

Therefore, a structured dashboard is required to consolidate project information into a single view. The dashboard should enable users to monitor key performance indicators, filter projects based on relevant categories, identify areas requiring attention, and support better management decisions.

3. PROJECT OBJECTIVES

The main objective of this project is to develop an interactive Excel dashboard for effective monitoring and analysis of elevator installation projects.

The specific objectives are:

- To create a centralized view of all elevator installation projects.
- To monitor project status, including completed and in-progress projects.
- To track installation stages and project handovers.
- To identify delayed projects and analyse delay status.
- To analyse project performance across different regions, customer types, and cities.
- To monitor project value and compare project values across different categories.
- To provide interactive filtering using Excel slicers.
- To present key project information through KPI cards, charts, and visualizations.
- To support management in identifying areas requiring attention and making better project-related decisions.

4. DATA OVERVIEW

The dataset used for this project contains information related to elevator installation projects. The data includes key project attributes required for monitoring project progress, status, delays, handovers, and project value.

The major fields considered in the analysis include:

- Project Status – Completed and In Progress
- Region – Central Kerala, North Kerala, and South Kerala
- Customer Type – Apartment, Commercial, Hospital, Hotel, Office, Residential, and Shopping Mall
- City – Project location
- Installation Stage – Different stages of elevator installation
- Handover Status – Completed and Pending
- Delay Status – Delayed, Early, In Progress, and On Time
- Project Value – Value associated with each project

The data was organized and analysed using Excel Pivot Tables and Pivot Charts. Interactive slicers were also used to filter the dashboard based on region, customer type, and project status.

5. DASHBOARD DEVELOPMENT

The dashboard was developed in Microsoft Excel using a structured project dataset and a combination of data analysis and visualization features.

The development process included organizing the project data into an Excel Table, creating supporting lists for data validation, and applying formulas for project-related calculations.

Pivot Tables were created to summarize project information from different perspectives. These Pivot Tables were then connected to Pivot Charts to provide visual representations of project status, installation stages, handovers, delays, customer types, locations, and project values.

KPI cards were developed to highlight important project metrics such as total projects, completed projects, projects in progress, handovers, delayed projects, on-time completion, average delay, and project value.

Interactive slicers were added for Region, Customer Type, and Status, allowing users to filter the dashboard dynamically and analyse project information based on selected categories.

The final dashboard combines these components into a single interactive interface designed for easy project monitoring and management reporting.

6. KEY PERFORMANCE INDICATORS (KPIs)

The dashboard uses KPI cards to provide a quick overview of the overall project performance. These indicators help users understand the current status of the project portfolio without having to review individual records.

The key performance indicators included in the dashboard are:

KPI	Value
Total Projects	305
Completed Projects	58
Projects In Progress	247
Handovers	58
Delayed Projects	92
On-Time Completion	45
Average Delay	1.97
Total Project Value	₹11,370.00
Average Project value	₹37.28

These KPIs provide a concise summary of project progress, delays, handover performance, and project value. The dashboard also allows users to interact with the KPIs through the available slicers.

7. DASHBOARD ANALYSIS & INSIGHTS

The dashboard provides multiple views of the elevator project portfolio through interactive charts and visual summaries.

The project status analysis shows that 58 projects are completed, while 247 projects are currently in progress. This provides a quick understanding of the overall project completion status.

The handover analysis helps monitor completed handovers and identify projects that may still require attention before final handover.

The installation stage analysis provides visibility into the current stage of each project and helps identify where projects are concentrated within the installation process.

The dashboard also provides region-wise and city-wise project analysis, allowing project distribution to be compared across different locations.

The customer type analysis shows the distribution of projects across categories such as Apartment, Commercial, Hospital, Hotel, Office, Residential, and Shopping Mall.

The delay analysis provides visibility into project delay status and helps identify projects that may require closer monitoring.

Finally, the project value analysis allows comparison of project values across regions, customer types, and project status, providing an additional financial perspective on the project portfolio.

The combination of these analyses provides a comprehensive view of project performance and supports more effective project monitoring.

8. KEY FINDINGS

The analysis performed through the dashboard provides the following key observations:

- The project portfolio contains 305 projects, of which 58 are completed and 247 are currently in progress.
- The dashboard records 58 handovers, providing a clear view of completed handover activity.
- 92 projects are classified as delayed, highlighting the importance of continuous delay monitoring.
- The dashboard records 45 on-time completions and an average delay of 1.97.
- Project distribution can be analysed across different regions, cities, and customer types.
- Project values can be compared across regions, customer types, and project status.
- Interactive slicers allow users to analyse the dashboard based on Region, Customer Type, and Status.

9. CONCLUSION

The Elevator Project Monitoring Dashboard demonstrates how Microsoft Excel can be used to organize, analyse, and visualize project information in an interactive and management-friendly format.

By combining KPI cards, Pivot Tables, Pivot Charts, and interactive slicers, the dashboard provides a centralized view of project progress, delays, handovers, installation stages, and project value.

The dashboard makes it easier to monitor multiple projects, identify areas requiring attention, and analyse project performance from different perspectives.

Overall, this project provided practical experience in Excel-based data analysis, data visualization, dashboard development, and interactive reporting.

10. TOOLS & TECHNOLOGIES USED

The following tools and Excel features were used to develop the project:

- Microsoft Excel – Data preparation, analysis, and dashboard development
- Excel Tables – Structured project data management
- Excel Formulas – Project calculations and data validation
- Pivot Tables – Data summarization and analysis
- Pivot Charts – Visual representation of project metrics
- Slicers – Interactive filtering of dashboard data
- Conditional Formatting – Visual identification of relevant project conditions
- Data Validation – Maintaining consistent data entry
- Dashboard Design – Combining KPIs, charts, and filters into an interactive reporting interface

Dataset Note: The dataset used in this portfolio project is simulated/sample data created for learning and demonstration purposes. It does not represent confidential company data.

11. DASHBOARD PREVIEW

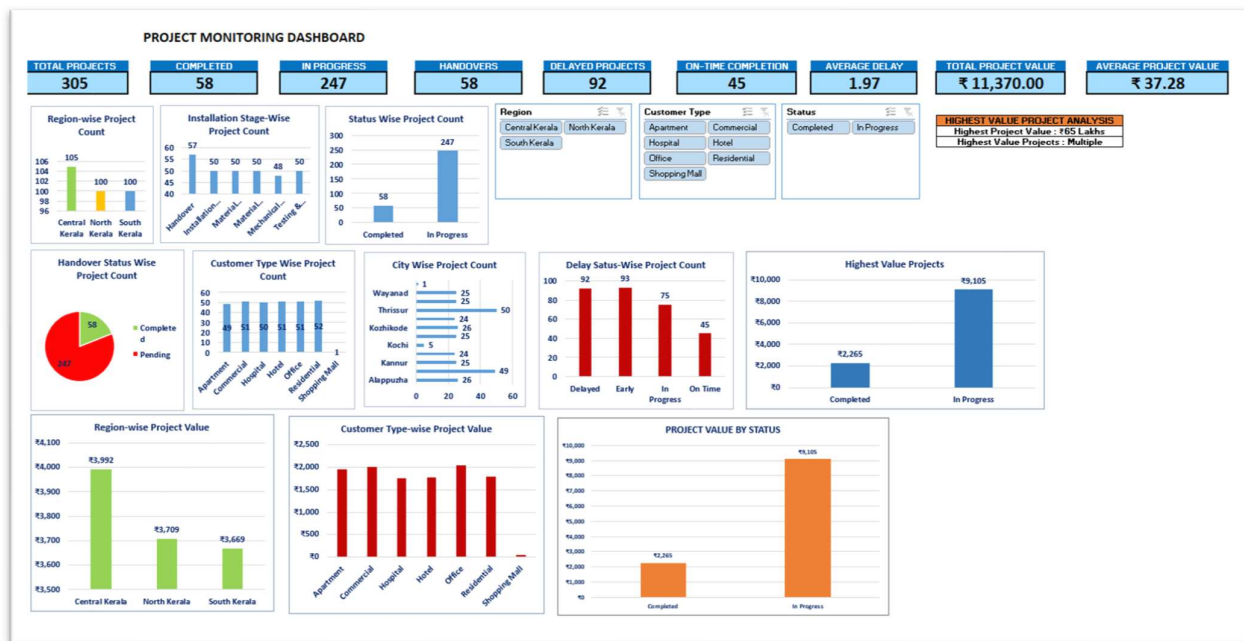


Figure 1: Elevator Project Monitoring dashboard

12. PROJECT REPOSITORY

The complete Excel dashboard workbook and project documentation are available in the GitHub repository.

<https://github.com/DenishaDenny/elevator-project-monitoring-dashboard>